

Sprint 027: Implementation of H3 Resolution Tier Boundary Validation in Spatial Monads

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Web of Life

Sprint 027 – Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

In spatially explicit thermodynamic ecosystem simulations, hierarchical spatial indexing structures dictate how biogeochemical stocks (Carbon, Nitrogen, Water) and energy fluxes are partitioned and conserved. Sprint 027 introduces a rigorous spatial resolution tier boundary validation mechanism (`isValidH3Resolution` and `assertValidH3Resolution`) within `src/spatial/h3_grid.ts`. This mechanism enforces strict domain boundaries ($\mathbb{Z} \cap [0, 15]$) across the Uber H3 hierarchical hexagonal grid, ensuring mass-balance invariance during monad refinements and preventing unphysical spatial recursion or floating-point discretization leakage.

1 Thermodynamic & Systems Ecology Foundation

The Web of Life simulation engine models planetary ecosystems as open thermodynamic systems driven by external solar inputs and constrained by fundamental physical laws:

1. **First Law of Thermodynamics (Mass Conservation):** When spatial monads undergo refinement (parent-to-children division) or compaction (children-to-parent aggregation), total matter stocks must remain invariant:

$$\sum_{i=1}^7 M_{\text{child},i} = M_{\text{parent}} \quad (1)$$

where $M \in \{C, H_2O, N, \text{minerals}\}$. Bounding resolution tiers prevents unallocated mass leaks or infinite spatial recursion.

2. **Second Law of Thermodynamics (Entropy & Granularity):** Finer resolution tiers (approaching tier 15, $\approx 0.9\text{m}$ edge length) capture localized microclimate entropy dissipation and trophic exchange, while coarser tiers (tier 0, $\approx 1107\text{km}$) aggregate macro-scale equilibrium. Boundary checks ensure thermodynamic matrices query valid physical scales.

2 Architectural & Code Implementation

Implemented within `src/spatial/h3_grid.ts`, the validation functions provide type-safe invariant checks:

```
export function isValidH3Resolution(resolution: number): boolean {  
  return Number.isInteger(resolution) && resolution >= 0 && resolution <= 15;  
}
```

```

export function assertValidH3Resolution(resolution: number): void {
  if (!isValidH3Resolution(resolution)) {
    throw new Error('Thermodynamic Invariant Violation: Invalid H3 resolution tier (${resolution})')
  }
}

```

3 Verification & Audit Matrix

Test Vector	Input (r)	Expected Result	Thermodynamic / Mass Implication
Lower Bound	0	true	Coarsest global planetary partition; valid macro-scale
Mid Tier	7	true	Regional watershed/biome scale; valid trophic cascades
Upper Bound	15	true	Finest local micro-habitat scale; valid organismal processes
Negative Out-of-Bounds	-1	false	Throws invariant error; prevents non-physical scales
Upper Out-of-Bounds	16	false	Throws invariant error; prevents infinite spatial partitioning
Floating-Point Injection	3.5	false	Throws invariant error; prevents fractional hexagonal cells

Table 1: Validation test vectors and corresponding thermodynamic implications for Sprint 027.

4 Conclusion

Sprint 027 successfully secures the spatial monad architecture by embedding immutable H3 resolution tier checks, safeguarding mass conservation across hierarchical scales.